# Leander Paul

Senior Backend Engineer — Identity & Banking Platforms

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## Summary

Senior backend engineer with 8+ years of experience across banking, identity and backend platforms. Currently building authorization services for Emirates NBD in Dubai, with production incident and disaster-recovery ownership. Previously worked on Zoho Accounts IAM.

## Experience

### Software Engineer — Sphere IT Consultants

Client: **Emirates NBD** · Dubai, UAE · Mar 2024 – Present

- Architected the retail app's OTP, IVR and Nafath authorization microservice, with database-configured validation checks between steps.
- Built manual review for partially eligible credit-card applicants and integrated credit-scoring and employment-verification services into the application journey.
- Own escalated production incidents on live banking services, from root-cause analysis to shipped fix.
- Led disaster-recovery environment setup for the team's services, meeting SAMA requirements.

**Technologies:** Java, Spring Boot, Node.js, Express

### Full Stack Engineer (Contract) — WTV Global

London, UK · Mar 2023 – Sep 2023

- Helped design APIs and three RabbitMQ-connected NestJS microservices for Capture, a carbon-footprint platform, and developed backend services.
- Established end-to-end tests with database reset and reseeding before each run; services ran in Docker containers.

### Software Engineer — Zoho Corporation

Chennai, India · Jan 2020 – Aug 2022

- Architected a hierarchical role-and-permission system for Zoho Accounts IAM and contributed to setting up SAML-based authorization.
- Built a proof of concept enabling LDAP queries against PostgreSQL through Apache Directory Server.
- Helped resolve Redis crashes caused by TTL-related CPU load by removing TTLs and balancing the load.
- Wrote migration scripts for sharded databases, set up and optimised an admin panel for inspecting and clearing user caches, and helped mentor junior engineers.

**Technologies:** Java, PostgreSQL, Redis

**Full Stack Engineer & Team Lead — Phosphene AI**
Chennai, India · Apr 2018 – Jan 2020

- Designed and built PixelView's image-upscaling pipeline using direct S3 uploads, Lambda and SQS, with on-demand compute that stopped after five idle minutes.
- Led three developers and built most of the React/Next.js frontend and Node.js backend, including Google integrations and Stripe payments.

**Software Engineer (Freelance)**
Chennai, India · Apr 2017 – Mar 2018

- Built and deployed two web applications on AWS EC2, including an e-commerce store with Razorpay payments and order/stock administration.

## Projects

**Shadow Platform** — personal project; not deployed to production
github.com/shadow-library/platform · 2024 – Present

- Built an OIDC/OAuth 2.1 identity provider with PKCE, refresh-token rotation and reuse detection, RBAC, SAML 2.0, passkeys and TOTP.
- Implemented token-family and session revocation on refresh-token reuse, plus back-channel logout through a database outbox with retries and dead-lettering.
- AI: LangGraph workflows with a judge-and-repair loop, calling models through Synapse, my AI gateway with failover and budgets.
- Built solo with AI-assisted tooling; 10,000+ automated tests across unit, integration and end-to-end suites.

Open source: @shadow-library npm packages, maintained in the platform monorepo.

## Skills

**Backend:** Java, Spring Boot, Node.js, TypeScript, Express, NestJS, Fastify, REST APIs, microservices

**Identity & security:** OAuth 2.0/2.1, OpenID Connect, SAML 2.0, WebAuthn/passkeys, TOTP MFA, RBAC, LDAP, OTP/IVR/Nafath authentication

**Data & messaging:** PostgreSQL, Redis, MongoDB, RabbitMQ, AWS SQS

**Cloud & delivery:** AWS (EC2, S3, Lambda, SQS), Docker, Kubernetes (k3d), GitOps, CI/CD (GitHub Actions)

## Education

**MSc Computer Science** — Queen Mary University of London — 2022 – 2023

**Bachelor of Technology** — St. Joseph's College of Engineering (Anna University), Chennai — 2016 – 2020